WORKED SOLUTION

STEP III 2018, Question 5

Pure • Arithmetic mean - geometric mean

STEP III 2018, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q5

The key idea

This is a proof of the AM–GM inequality by induction, but with a twist: instead of showing $A_n \geq G_n$ directly, we show that the weighted gap $n(A_n - G_n)$ can never decrease when a new number is added. The gap starts at zero when $n = 1$, so it stays non-negative. The quantity λ_k is chosen so that G_{k+1} becomes a simple multiple of G_k, which turns the whole comparison into a one-variable polynomial inequality.

Solution

Part (i)

Two facts link step k to step $k+1$. For the sums,

$$(k+1)A_{k+1} = a_1 + \cdots + a_k + a_{k+1} = kA_k + a_{k+1}.$$

For the products, $G_k^k = a_1 a_2 \cdots a_k$, and the definition of λ_k gives $a_{k+1} = G_k \lambda_k^{k+1}$. So

$$G_{k+1}^{k+1} = G_k^k \, a_{k+1} = G_k^k \cdot G_k \lambda_k^{k+1} = (G_k \lambda_k)^{k+1}, \qquad \text{so} \qquad G_{k+1} = G_k \lambda_k,$$

taking positive $(k+1)$th roots, since $G_k > 0$ and $\lambda_k > 0$. This is exactly why λ_k was defined as it was.

Now we compare the two sides:

$$\begin{aligned}(k+1)(A_{k+1} - G_{k+1}) - k(A_k - G_k) &= kA_k + a_{k+1} - (k+1)G_{k+1} - kA_k + kG_k \\ &= G_k \lambda_k^{k+1} - (k+1)G_k \lambda_k + kG_k \\ &= G_k(\lambda_k^{k+1} - (k+1)\lambda_k + k).\end{aligned}$$

Since $G_k > 0$, the left-hand side is ≥ 0 if and only if the bracket is ≥ 0. This is the required equivalence.

Part (ii)

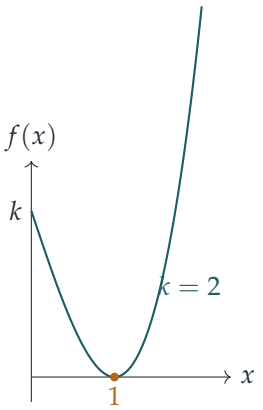
Differentiating,

$$f'(x) = (k+1)x^k - (k+1) = (k+1)(x^k - 1).$$

For $0 < x < 1$ we have $x^k < 1$, so $f'(x) < 0$, and for $x > 1$ we have $x^k > 1$, so $f'(x) > 0$. Hence f is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$, and its least value on $x > 0$ is

$$f(1) = 1 - (k+1) + k = 0.$$

So $f(x) \geq 0$ for all $x > 0$. Moreover, if $x \neq 1$ then $f(x) > f(1) = 0$ by the strict monotonicity on either side of 1. So $f(x) = 0$ only when $x = 1$, and $f(1) = 0$ gives the converse.



(For $k = 2$, $f(x) = x^3 - 3x + 2 = (x-1)^2(x+2)$; the double root at $x = 1$ is where the graph just touches the axis.)

Part (iii)

Write $d_n = n(A_n - G_n)$ for each positive integer n.

(a) By part (ii) with $x = \lambda_k > 0$, we have $\lambda_k^{k+1} - (k+1)\lambda_k + k \geq 0$, so by part (i)

$$d_{k+1} \geq d_k \qquad \text{for every } k \geq 1.$$

Also $A_1 = a_1 = G_1$, so $d_1 = 0$. Hence

$$0 = d_1 \leq d_2 \leq d_3 \leq \cdots \leq d_n,$$

so $d_n \geq 0$, and dividing by $n > 0$ gives $A_n \geq G_n$ for all n.

(b)

Ask yourself If $A_n = G_n$, what does that tell us about the earlier steps of the chain, not just the last one?

If $A_n = G_n$ then $d_n = 0$, and the chain becomes $0 = d_1 \le d_2 \le \cdots \le d_n = 0$. A non-decreasing chain that starts and ends at 0 must be constant, so $d_{k+1} = d_k$ for each $k = 1, \ldots, n-1$. By the identity in part (i) this means $G_k f(\lambda_k) = 0$, and as $G_k > 0$, $f(\lambda_k) = 0$. By part (ii), $\lambda_k = 1$, which says $a_{k+1} = G_k$, for each $k = 1, \ldots, n-1$.

Now we argue by induction on k. For $k = 1$: $a_2 = G_1 = a_1$. If $a_1 = \cdots = a_k = a$, then $G_k = (a^k)^{1/k} = a$, so $a_{k+1} = G_k = a$. Hence $a_1 = a_2 = \cdots = a_n$, as required.

Checks

Take $a_1, a_2, a_3 = 1, 4, 16$. Then $A_2 = 2.5$, $G_2 = 2$, $A_3 = 7$, $G_3 = 4$, so $d_1, d_2, d_3 = 0, 1, 9$: increasing, as promised. Here $\lambda_2 = (16/2)^{1/3} = 2$, and $G_2 f(\lambda_2) = 2(8 - 6 + 2) = 8 = d_3 - d_2$, which confirms the identity from part (i). A computer check of the identity and of $d_{k+1} \ge d_k$ on many thousands of random positive sequences found no exceptions.

Common slips

- Proving only one direction in part (i). Writing the difference as $G_k f(\lambda_k)$ gives both directions at once, because $G_k > 0$.
- In (iii)(b), concluding only that $\lambda_{n-1} = 1$. You need every step of the chain to be an equality, which comes from the chain starting at $d_1 = 0$.
- Forgetting to say why $\lambda_k > 0$ before applying part (ii), which only covers $x > 0$.

Beyond STEP

The inequality $x^{k+1} - (k+1)x + k \ge 0$ is a special case of Bernoulli's inequality, and AM–GM itself is a special case of Jensen's inequality for the concave function $\ln x$. These convexity inequalities are basic tools in first-year analysis. Several other proofs of AM–GM, including Cauchy's "forward–backward" induction, are a good exercise; Bartle and Sherbert, *Introduction to Real Analysis*, covers the analysis background.

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